COMPOUNDING

The Rule Of 72, And The One Place It Lies To You

Divide 72 by the rate, get the doubling time. The most useful piece of mental arithmetic in finance, and wrong in a direction that matters. Where it is safe, where it drifts, and what to use when the number leaves your head and enters a plan.

CONTENTS

01 The rule

Divide 72 by the growth rate and you get how long it takes to double. That is the whole rule.

- **10 percent a month.** 72 divided by 10 is **7.2 months** to double.
- **8 percent a year.** 72 divided by 8 is **9 years**.
- **It runs backwards too.** Doubled in 6 periods? Then you grew at roughly 72 divided by 6, or **12 percent** a period.

> THINK OF IT AS A SCALE BAR
>
> A map's scale bar is accurate in the middle of the sheet and drifts at the edges, and you use it anyway because it is faster than measuring. **The Rule of 72 is that scale bar for growth.**

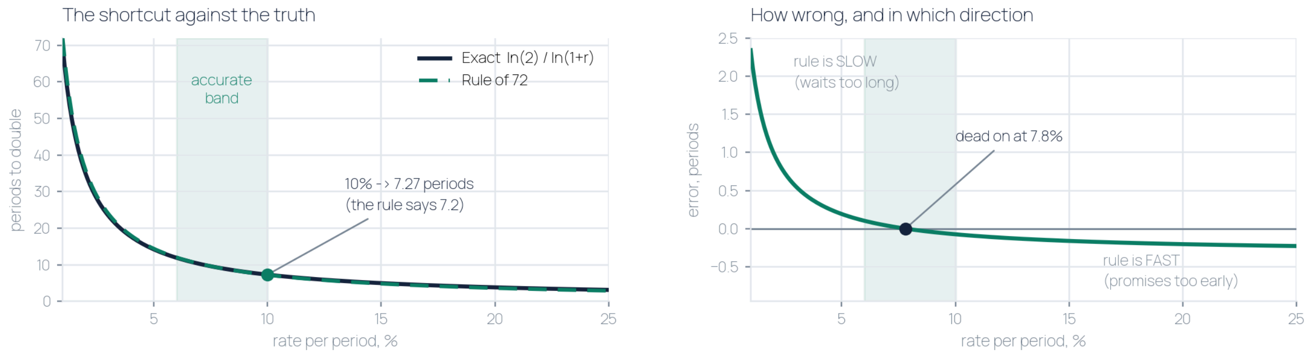
02 Why it works

The rule is not folklore. It is a real formula with the ugly parts rounded off.

- **The true answer.** Periods to double equals **ln(2) divided by ln(1 plus r)**. (ln: the natural logarithm, the function that turns compounding into simple addition)
- **ln(2) is 0.693, the doubling constant.** And for small rates ln(1 plus r) is very close to r, so the answer is roughly **69.3 divided by the rate in percent**.
- **So why 72 and not 69.3?** Two reasons, and the second is the real one. It nudges the estimate toward accuracy at the rates people actually use, and **72 divides cleanly by 2, 3, 4, 6, 8, 9 and 12**. 69.3 divides by nothing.

03 Where it drifts

The rule has a sweet spot. Inside it the error is invisible. Outside it the error has a direction.



Left: the shortcut against the exact formula, 1 to 25 percent. Right: the signed error. Above the line the rule is slow and makes you wait; below it the rule is fast and promises a double you have not reached. Shaded band is 6 to 10 percent.

- **It is exactly right at about 7.8 percent.** That is the crossover, and it is the only rate where the shortcut and the truth agree.
- **Below it runs slow, above it runs fast.** At 1 percent the rule says 72 when the answer is 69.7, handing you **2.3 extra periods** you do not owe. At 20 percent it says 3.6 when the answer is 3.8, promising a double **0.2 of a period early**.
- **Between 6 and 10 percent the error is under a tenth of a period.** That is the band it was tuned for and where you should trust it without thinking.

04 The catch

Here is the rule failing in a small, honest way, on a real calculator.

- **Type 1.1 to the power 7.2.** You get **1.986220**, not 2. The rule sent you to 7.2 and 7.2 is not enough.
- **You land 0.69 percent short of a double.** Small. Invisible on one pass.
- **The true doubling point at 10 percent is 7.2725 periods.** Just over seven and a quarter, not seven and a fifth.
- **The fix costs nothing.** Use the rule to decide whether something is worth modelling. Use **ln(2) divided by ln(1 plus r)** when the number goes in a plan.

THE LINE TO KEEP

The Rule of 72 is a **decision tool, not a projection tool**. It is for the moment you are deciding whether a number is worth taking seriously. Once you have decided it is, put the shortcut down and use the formula.

SOURCES & NOTE

Sources: The earliest known statement is in Luca Pacioli's Summa de Arithmetica, 1494, which gives the rule without deriving it; The exact relation is ln(2) divided by ln(1 plus r). ln(2) is 0.6931; Every figure in this guide was computed by 6-Scripts/build_rule72_chart.py, which prints the values it draws; The 1.1 to the power 7.2 case is Mann's own, on his calculator, August 22nd 2026.

Educational, not financial advice. The arithmetic is settled. What anyone compounds is not.

Glossary

Rule of 72 Divide 72 by the growth rate per period to estimate periods to double.

Compounding Each period grows the result of the last one, not the original amount.

ln natural logarithm The function that turns repeated multiplying into simple adding.

ln(2) 0.6931. The doubling constant, and the reason the rule uses a number near 70.

Crossover About 7.8 percent, the one rate where the rule is exactly right.

Period Whatever unit the rate is quoted in. Months if monthly, years if annual.

Rule of 70 The same shortcut using 70. More accurate at low rates, harder to divide.

Rule of 69.3 The mathematically pure version. Most accurate, least usable in your head.